$$a_1 = f(z_1) = \frac{e^{z_1}}{\sum e^{z_i}} \quad 0 + 1 \quad \underline{\underline{0 + \infty}}$$

$$a_2 = f(z_2) = \frac{e^{z_2}}{\sum e^{z_i}} \quad 0 + 1$$

$$a_3 = f(z_3) \qquad \vdots$$

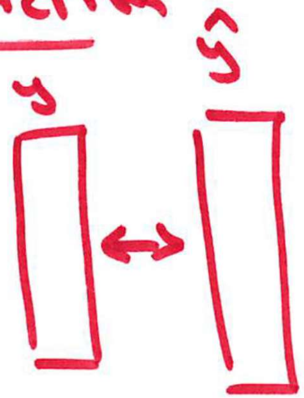
$$a_4 = f(z_4) \qquad \vdots$$

$$a_5 = f(z_5) \qquad \vdots$$

$$\frac{\cancel{\sum e^{z_i}}}{\cancel{\sum e^{z_i}}} \; 1$$

$$e^{z_1} \quad e^{z_2} \quad \ldots\ldots \quad e^{z_5}$$

Loss Function



Loss func → distance measure

↓

a 'norm'

$\frac{1}{N}\sum |y_i - \hat{y}_i|$

MAE

SAE

MAD

$\frac{1}{N}\sum (y_i - \hat{y}_i)^2$

↓

SSE

MSE

RMSE

Reg

Cross entropy

binary classificat.

Cross entropy

$$L = -\left(y_i \ln(\hat{y}_i) + (1-y_i)\ln(1-\hat{y}_i) \right)$$

$$L = \begin{cases} -\ln \hat{y}_i & \text{if } y_i = 1 \\ -\ln(1-\hat{y}_i) & \text{if } y_i = 0 \end{cases}$$

truth	pred
1	0
0	1
1	0.9
1	0.4

$(\hat{y} - y)^2$

→ Reg → not a big deal cause you are 1 off. $\underline{(1-0)^2}$

bin Class → $-\ln(0) \to \infty$

$\cancel{(\hat{y} - y)^2}$ $-\ln(1-1) \to \infty$